

2021

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer from **all** the Groups as directed.*

Group – A

(Objective Type Questions)

1. Choose the correct answer in each of the following :

1×5 = 5

(a) OSI stands for :

- (i) Open System Inter connection
- (ii) Open System Internet connect
- (iii) Open System Intranet connect
- (iv) None of these

(b) POP stands for :

- (i) Pre Office Protocol

- (ii) Post Office Protocol
 - (iii) Protocol of Post
 - (iv) None of these
- (c) What is the number of layers in the OSI model ?
- (i) 2 layers
 - (ii) 4 layers
 - (iii) 7 layers
 - (iv) 9 layers
- (d) Identify the layer which provides services to the user :
- (i) Session layer
 - (ii) Application layer
 - (iii) Presentation error
 - (iv) Physical layer
- (e) FDM stands for ;
- (i) Frequency Division Multiplexing
 - (ii) Frequency Division Multiple
 - (iii) Frequency Division Motion
 - (iv) None of these

2. Fill in the blanks :

$$1 \times 5 = 5$$

- (a) The network, which interconnects Multiple Station over a very small area is called

PPP

- (b) A Protocol is set of rules that controls the data communication.
- (c) _____ layer takes raw transmission facility and transform it in to line.
- (d) The Network layer is concerned with the controlling of operation of the subnet.
- (e) IP is the basic protocol of the Internet.

Part - B

(Short-answer Type Questions)

Answer any four of the following : $5 \times 4 = 20$

3. (a) Explain the types of Network topologies with diagram.
- (b) Differentiate between Analog and digital signal.
- (c) Explain the different types of protocol in Computer Network.
- (d) Explain the following :
- (i) Repeater
 - (ii) Router
 - (iii) Gateway
 - (iv) HUB

- (e) What is Pulse Code Modulation (PCM) ?
- (f) Explain the stop and wait ARQ.

Part – C

(Long-answer Type Questions)

Answer any four questions of the following :

$10 \times 4 = 40$

4. (a) Differentiate between Circuit Switching and Packet Switching.
- (b) What is OSI Model ? Describe the different layers of OSI Model.
- (c) Explain the Error Detection and Correction of the Data Link Layer.
- (d) Explain the Handshaking Method of transport layer.
- (e) Describe the Error and Flow Control in transport layer.
- (f) Explain the following :
- (i) Routing
 - (ii) CSMA / CD Protocol
 - (iii) HTTP

